

DBSCAN Density Clustering: Numerical Walkthrough

DBSCAN groups points based on local density. It does not require you to pre-specify the number of clusters, and it is capable of discovering clusters of arbitrary shapes (like concentric rings) while cleanly filtering out outliers. It relies on a search radius ϵ (**Epsilon**) and a neighborhood threshold **MinPts**.

The Sample Dataset

Imagine we have 5 points plotted on a 2D coordinate grid:

- **Point A:** (1, 1)
- **Point B:** (1, 2)
- **Point C:** (2, 1)
- **Point D:** (1, 10)
- **Point E:** (10, 10)

We set $\epsilon = 3$ and **MinPts** = 2.

Step-by-Step Execution

Round 1: Scanning Point A (1, 1)

1. Select Point A as our start point.
2. Scan neighbors within distance $\epsilon = 3$: B (1, 2) and C (2, 1) are inside. D and E are outside.
3. Count neighborhood size: Neighbors of A are {A, B, C} (3 points).
4. Evaluate density: Since the count (3) is $\geq$ MinPts (2), **Point A is labeled a Core Point**.
5. Form cluster: A starts **Cluster 1**. Its unvisited neighbors {B, C} are added to the expansion queue.

Round 2: Expanding via Point B (1, 2)

1. Pop Point B from the queue.

- 2. Scan neighbors: Neighbors are {A, B, C} (3 points).
- 3. Evaluate density: Since count $\geq$ MinPts, **Point B is also a Core Point**.
- 4. Expand: B joins Cluster 1.

Round 3: Expanding via Point C (2, 1)

- 1. Pop Point C from the queue.
- 2. Scan neighbors: Neighbors are {A, B, C} (3 points).
- 3. Evaluate density: Since count $\geq$ MinPts, **Point C is a Core Point**.
- 4. Expand: C joins Cluster 1. The queue is now empty. Cluster 1 is complete!

Round 4: Scanning Point D (1, 10)

- 1. Select Point D (next unvisited point).
- 2. Scan neighbors: The only neighbor of D is {D} (1 point). Distance to E (10, 10) is 9.0 (Outside).
- 3. Evaluate density: Since the count (1) is $<$ MinPts (2), Point D is not a Core Point.
- 4. Label status: Since D has no connections to Core points, it is labeled as **Noise (Outlier)**.

Round 5: Scanning Point E (10, 10)

- 1. Select Point E (last unvisited point).
- 2. Scan neighbors: Only {E} (1 point).
- 3. Evaluate density: Since count $<$ MinPts, Point E is not a Core Point.
- 4. Label status: Point E is labeled as **Noise (Outlier)**.

Point	Coordinates	Neighbors Found	Class Type	Final Assignment
Point A	(1, 1)	3	Core Point	Cluster 1
Point B	(1, 2)	3	Core Point	Cluster 1
Point C	(2, 1)	3	Core Point	Cluster 1
Point D	(1, 10)	1	Outlier	Noise (Isolated Outlier)
Point E	(10, 10)	1	Outlier	Noise (Isolated Outlier)

Unsupervised Learning Masterclass • © 2026 Visual Explorer Project